

Tutorial on ProSpecTool: expert spectra preprocessing toolbox for PLS

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Introduction

The present protocol provides general guidelines for users working with the ProSpecTool MATLAB toolbox; an objective and automatic system for preprocessing method optimisation of spectroscopic data for the consequent prediction based on Partial Least Squares (PLS) regression algorithm. This tool allows users to perform and test many preprocessing methods of vibrational spectroscopic data (e.g. UV-Vis, NIR, MIR and Raman) and helps choose the optimal one.

ProSpecTool Initial Window

0. **Accessing ProSpecTool:** To use the ProSpecTool toolbox, add the folder where the downloaded files are to your MATLAB pathways and enter either "ProSpecTool" or "PST" in the Command Window.

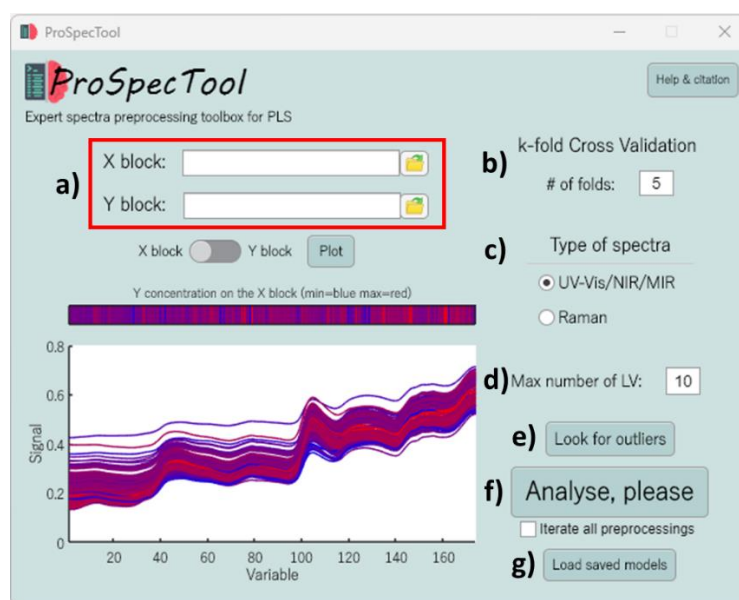


Figure 1. Main window of the ProSpecTool toolbox.

1. **Loading Data:** On the initial tab the **X**-block (spectroscopic data) and the **Y**-block (reference data) are loaded (Figure 1). Load data from the MATLAB workspace by entering the array name (double format) or by clicking the folder icon to load a .mat file (Figure 1a) where the array is saved. Ensure the data is organised with samples as rows and variables as columns.
2. **Data Visualisation:** Check that the data is loaded correctly by plotting both the **X**- and **Y**-blocks (refer to Figure 2a and b). The **X**-block will be color-coded based on the **Y**-block values, helping explore spectroscopic regions related to the dependant variable. The **Y**-block displays a histogram with a best fit Gaussian curve to visualize data distribution and gaps.

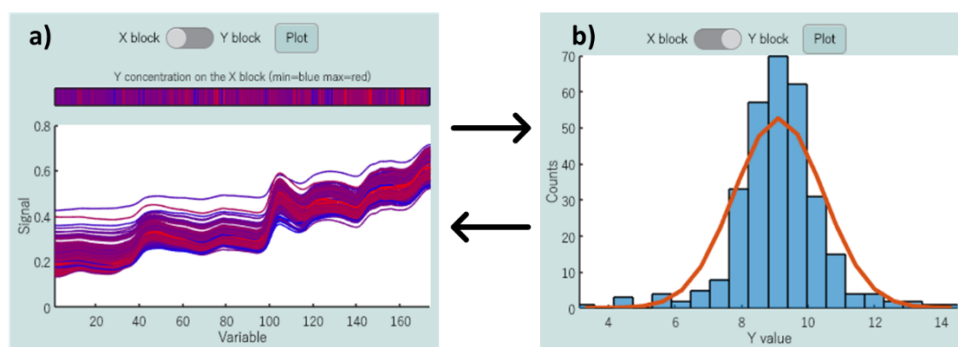


Figure 2. Graphical representation of X- and Y-blocks.

3. **Cross-Validation Setup:** Specify the number of folds for a random subsets cross-validation (Figure 1b), with a default value of five.
Note: Caution with structured datasets, as over-optimistic validations may lead to misleading conclusions.
4. **Data Type Selection:** Choose the data type from UV-Vis/NIR/MIR or Raman spectra categories (Figure 1c). The selection influences the pre-processing combinations applied to the spectra.
5. **Number of Latent Variables (LV):** Decide the maximum number of LVs to be calculated, from which the optimal number of LVs will be selected for the outlier detection and all subsequent models (Figure 1d).
6. **Outlier Detection:** Click "Look for outliers" (Figure 1e) to calculate a PLS regression model using raw spectra. Further details on this process will be covered in the following section.
7. **Analysing Data:** Initiate the analysis by clicking "Analyse, please" (Figure 1f). This step evaluates statistical parameters, noise, and the presence of multiplicative/ additive effects in the spectra to select which preprocessings to iterate.
All possible combinations can be calculated by ticking the "Iterate all preprocessings" button. Results will be discussed in subsequent sections.
Note: When this button is clicked, the model_struct and Report in the Workspace will be rewritten, please save them before further analysis.
8. **Load Previous Model:** If the toolbox has been used before and results were saved, they can be loaded for further analysis of preprocessing results (Figure 1g) from a file containing "model_struct" and "Report" in a .mat file.

Look for outliers Tab

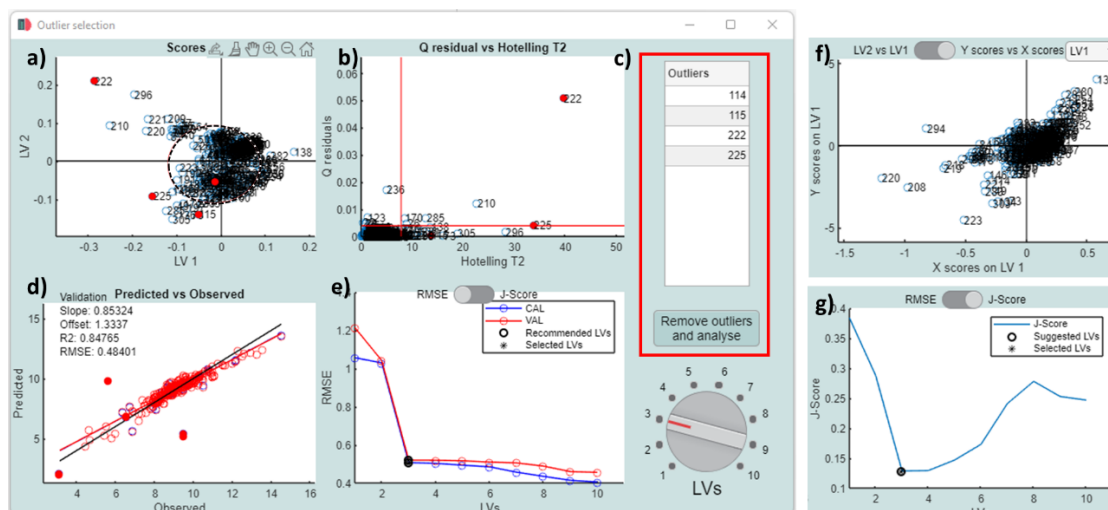


Figure 3. Outlier selection window, where a PLSR model is fitted to the raw data.

1. **Score plot:** Explore tendencies in the spectra by observing a representation of the first two LVs of the **X**-block (LV2 versus LV1) in a Score plot (Figure 3a).
2. **Score plot from Y and X:** Switch to a visual representation of the Scores of the **Y**-block versus the scores of the **X**-block within the same graph as the Score plot (Figure 3f). Use the drop-down option to select the latent variable for both blocks. A linear trend between the scores of both blocks represents a useful LV for the regression.
3. **Influence plot:** Analyse the representation of Q residuals versus Hotelling's T^2 in the Influence plot (Figure 3b). The brush tool can be used on this plot to select possible outliers. The selected outliers will be marked in red in other graphs within this tab, and listed in the Outliers selection.
4. **Outliers selection:** List of the selected outliers by their row number in the input (Figure 3c). Users may choose to remove selected outliers from the data matrix for further analysis by clicking "Remove Outliers and Analyse," without making any changes to the original data matrix in the MATLAB workspace. This will lead to step 7 of the previous tab without the selected outliers.
5. **Predicted versus observed values:** Assess the prediction performance of raw spectra and obtain performance parameters of the PLS regression model (Figure 3d). Calibration is represented as blue circles, while cross-validation is depicted as red circles.
6. **RMSE curve:** Visualise the Root Mean Square Error (RMSE) of calibration and cross-validation by selecting the number of latent variables for the model (Figure 3e). This selection before the detection of outliers is necessary as the outliers may change with the selected number of LVs.
7. **J-Score curve:** Switch to a visual representation of the evolution of the J-Score (a joint model performance parameter) in the same graph as the RMSE (Figure 3g). The J-Score aids in selecting the optimal number of latent variables, as explained in:
Ezenarro J., Schorn-García D., Aceña L., Mestres M., Busto O., Boqué R. (2023) J-Score: A new joint parameter for PLSR model performance evaluation of spectroscopic data,

8. **LVs selection tool:** Change the number of latent variables used by the PLS regression algorithm by using the LVs selection tool (Figure 3f). Each of the presented graphs will dynamically adjust accordingly.

Preprocessed PLS models tab

0. **Start the analysis:** When the 'Analyse, please' (Figure 1f) or 'Remove outliers and analyse' (Figure 3c) buttons are clicked, a wait bar is shown, representing the number of models calculated (which will vary based on the characteristics of the data).

Model	Smoothing	S. Window	Pol. Order	Derivative	Normalisation	N. Pol. Order	Limit	Autoscale	Sugg. LVs (J-Score)	J-Score	RMSE-CV	R2-CV	RV Noise
1	No				0 No			No		3	0.1262	0.3846	0.9159
2	No				0 SNV			No		3	0.1409	0.4126	0.9042
3	No				0 Baseline			No		3	0.1349	0.4113	0.9039
4	No				0 Detrending	1		No		4	0.1430	0.3936	0.9120
5	No				0 Detrending	2		No		3	0.1594	0.4208	0.8944
6	Gaussian	7			0 No			No		3	0.1255	0.3859	0.9154
7	Gaussian	7			0 SNV			No		3	0.1359	0.4144	0.9038
8	Gaussian	7			0 Baseline			No		3	0.1310	0.4117	0.9037
9	Gaussian	7			0 Detrending	1		No		4	0.1335	0.3930	0.9123
10	Gaussian	7			0 Detrending	2		No		3	0.1524	0.4366	0.8917
11	Gaussian	9			0 No			No		3	0.1222	0.3837	0.9163
12	Gaussian	9			0 SNV			No		3	0.1280	0.4086	0.9056
13	Gaussian	9			0 Baseline			No		3	0.1329	0.4157	0.9019
14	Gaussian	9			0 Detrending	1		No		3	0.1303	0.4104	0.9043
15	Gaussian	9			0 Detrending	2		No		3	0.1523	0.4413	0.8893
16	Gaussian	11			0 No			No		3	0.1196	0.3823	0.9169
17	Gaussian	11			0 SNV			No		3	0.1247	0.4068	0.9050
18	Gaussian	11			0 Baseline			No		3	0.1277	0.4102	0.9043
19	Gaussian	11			0 Detrending	1		No		4	0.1300	0.3947	0.9115
20	Gaussian	11			0 Detrending	2		No		3	0.1497	0.4437	0.8881

Figure 4. Table describing the PLSR models obtained from the preprocessed datasets.

1. **Preprocessed PLS models:** A table showing all the models calculated, the preprocessing methods (and combinations) used for them, and the figures of merit (Figure 4). Additionally, in the Workspace, a 'model_struct' and 'Report' are saved.
 - a. **model_struct:** The information (data, preprocessing method and figures of merit) about all the calculated PLS models is stored in this structure.
 - b. **Report:** The information about the input data is stored here, and the spectral characteristics on which the iterated preprocessing methods are selected.

The showed models may differ from the number of models calculated, since there is a filtering tool (explained in the following subsection).

The user may order the models by any of the columns (from smaller to bigger, or alphabetically in the preprocessing columns that provide a text).

To open up a model, the user needs to click on the Model number, that will recalculate the model in that moment, opening a new tab (discussed in the following section). Inconsistencies in the figures of merit between the table and the recalculated model may indicate that the validation is not robust enough.

Note: The same model cannot be opened up twice one after another, a different model must be opened in between.

2. **Filtering tool:** A small table with three tabs will open up together with the table, with some default values:

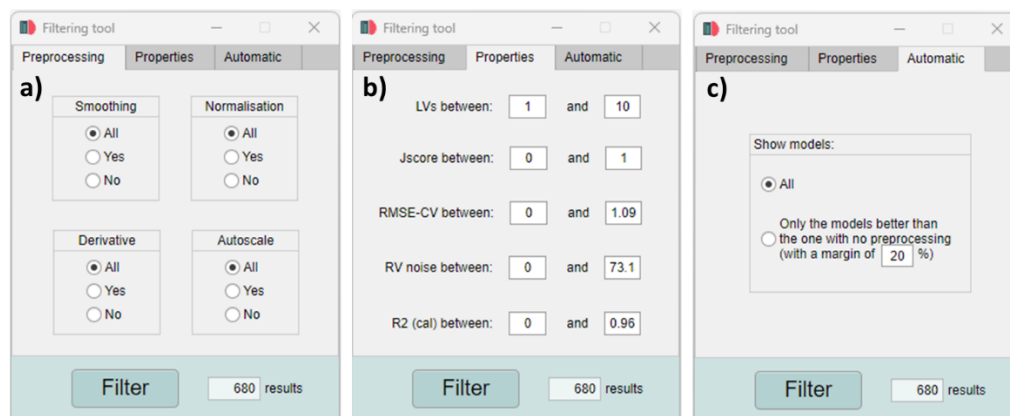


Figure 5. Tool that enables the user to filter the created models.

- In the preprocessing tab, the preprocessing types are marked as 'All', meaning that all models will be shown. The user may select to 'Yes' or 'No' to show only the models that use them or do not use them, respectively, based on their aim (Figure 5a).
- In the properties tab, the user may adjust the range of values for the figures of merit to filter the best models by various parameters at the same time (Figure 5b).
- In the automatic tab, the previous step can be done by comparing the models with the performance of the model with no preprocessing (Figure 5c).

Model visualisation tab

When a model is opened from the previous tab, the following characteristics are shown:

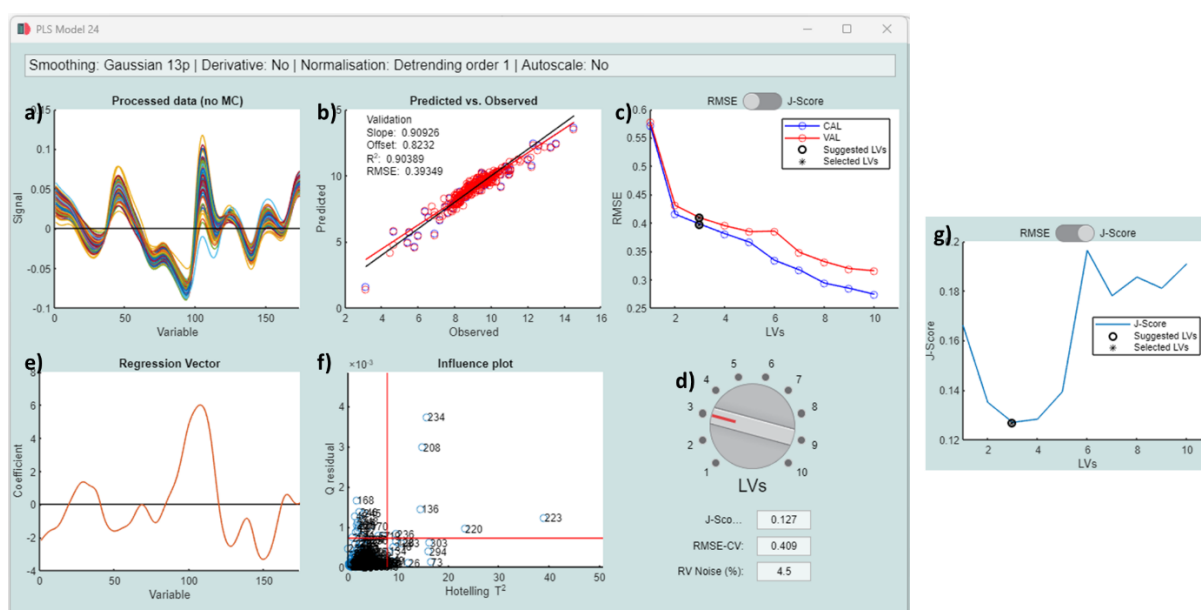


Figure 6. Window of the descriptive plots of a resulting PLSR model.

- Spectra plot:** The preprocessed spectra are plotted (before mean centring, for interpretability) (Figure 6a).

2. **Predicted versus observed values:** Assess the prediction performance of preprocessed spectra and obtain performance parameters of the PLS regression model (Figure 6b). Calibration is represented as blue circles, while cross-validation is depicted as red circles.
3. **RMSE curve:** Visualise the Root Mean Square Error (RMSE) of calibration and cross-validation by selecting the number of latent variables for the model (Figure 6c). This and the following plot can help the user making sure the selected number of LVs is the optimal one.
4. **J-Score curve:** Switch to a visual representation of the evolution of the J-Score (a joint model performance parameter) in the same graph as the RMSE (Figure 6g). The J-Score aids in selecting the optimal number of latent variables, as explained in:
Ezenarro J., Schorn-García D., Aceña L., Mestres M., Busto O., Boqué R. (2023) J-Score: A new joint parameter for PLSR model performance evaluation of spectroscopic data, Chemometrics and Intelligent Laboratory Systems, 240, 104883, 10.1016/j.chemolab.2023.104883.
5. **LVs selection tool:** Change the number of latent variables used by the PLS regression algorithm by using the LVs selection tool (Figure 6d). Each of the presented graphs will dynamically adjust accordingly.
6. **Regression vector plot:** The regression vector of the PLSR model is shown, remarking the most important variables for the prediction (Figure 6e).
7. **Influence plot:** Analyse the representation of Q residuals versus Hotelling's T^2 in the Influence plot (Figure 8f). The brush tool can be used on this plot to select possible outliers. The selected outliers will be marked in red in other graphs within this tab, and listed in the Outliers selection.